Lesson-End Project

Implementing Web App

Project agenda: To implement web app for hosting a website for your business

Description: Edu-sim an educational technology company, is evaluating Azure web apps to migrate its PHP websites from on-premises Windows servers. The IT team assesses the current setup, checks Azure compatibility, and plans the migration using Azure Migrate. They set up and secure Azure web apps, establish CI/CD pipelines with Azure DevOps, configure deployment slots, and integrate automated testing. After functional, performance, and user acceptance testing, the team deploys to production and implements continuous monitoring.

Tools required: Azure account with administrator access

Prerequisites: Resource group

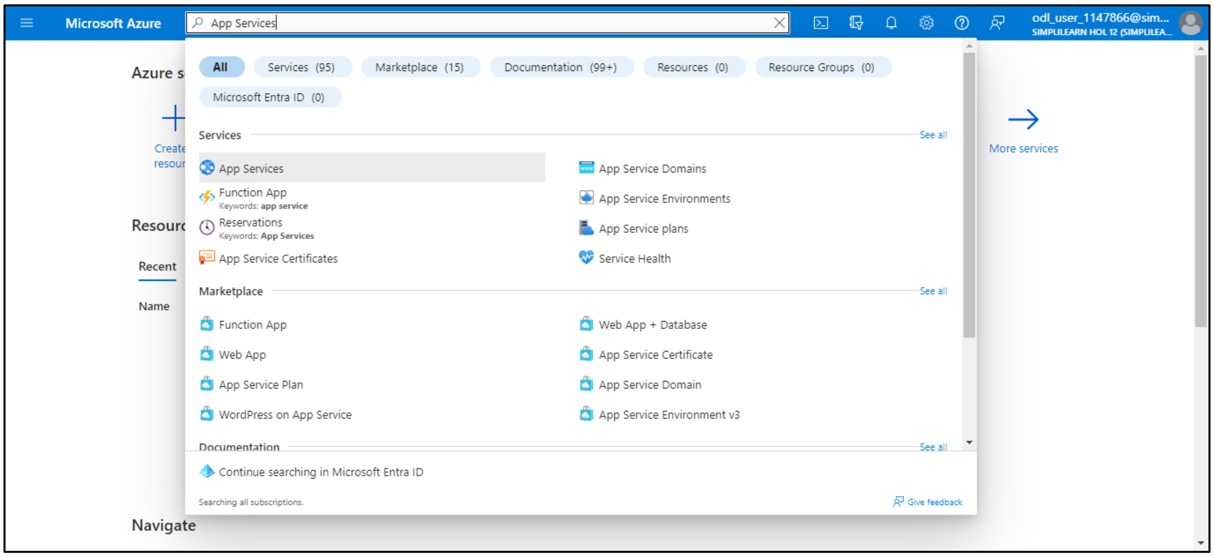
Expected deliverables: Add screenshots of configuring and deploying the web apps and demonstrate the deployment of code to the staging deployment slot

Steps to be followed:

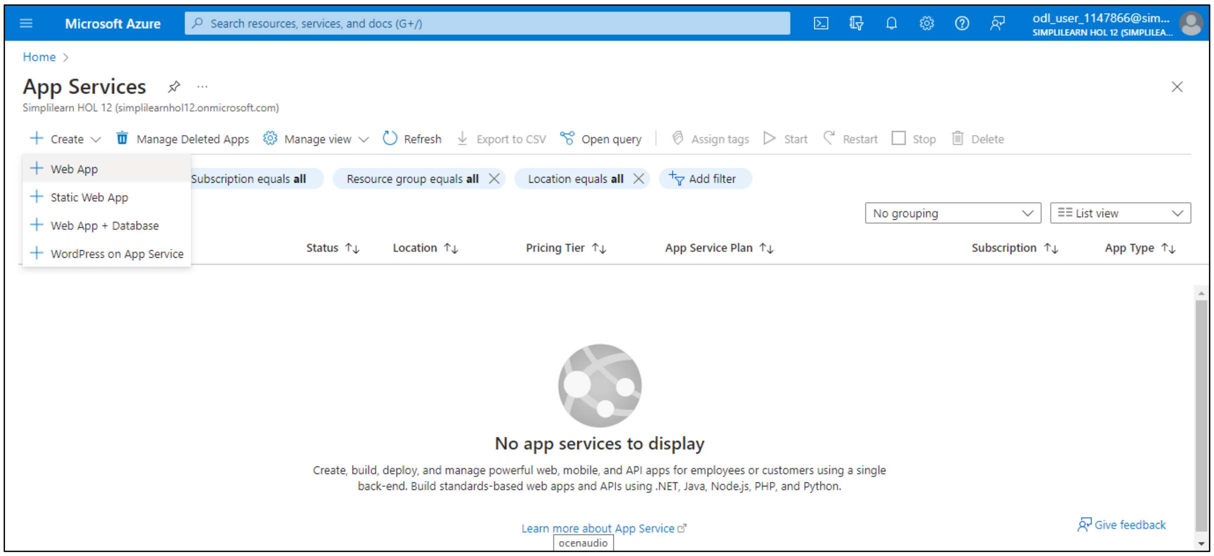
1. Create an Azure web app
2. Create a staging deployment slot
3. Configure web app deployment settings
4. Deploy code to the staging deployment slot

Step 1: Create an Azure web app

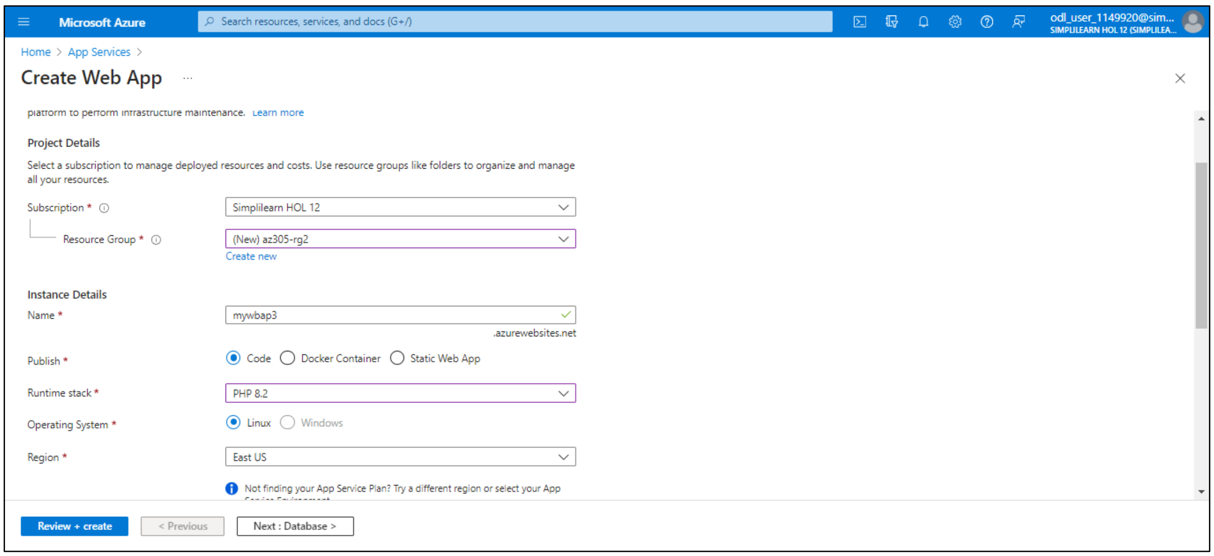
1.1 Log in to the **Azure portal**; search for and select **App Services**



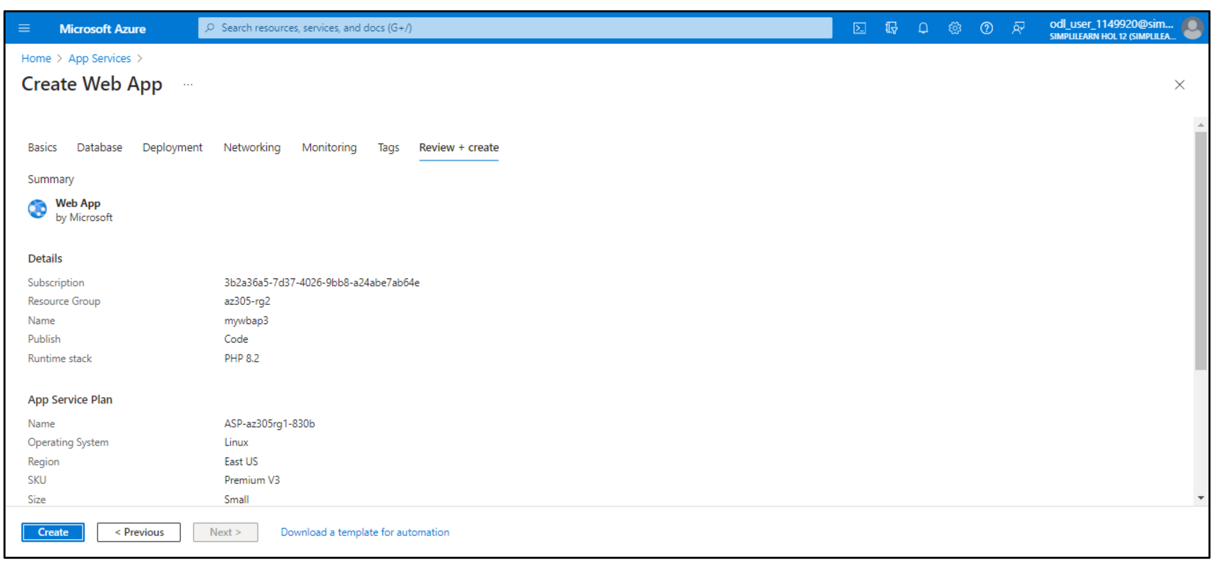
1.2 In the **App Services** page, click on **+Create** and select **Web App**



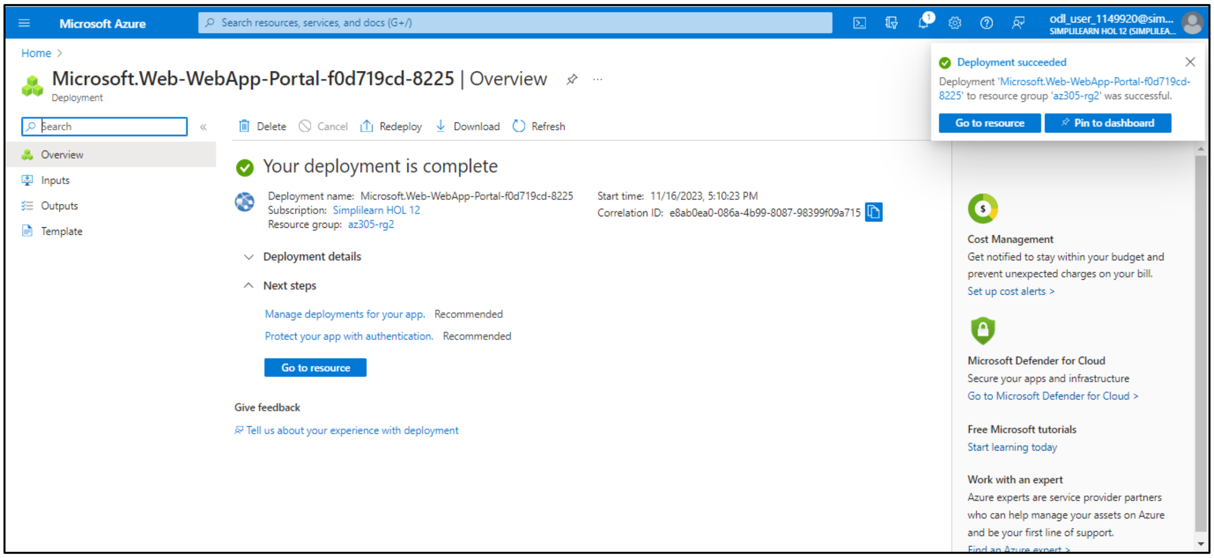
1.3 In the **Create Web App** page, under **Project Details**, specify the information as shown in the screenshot below and click on **Review + create**



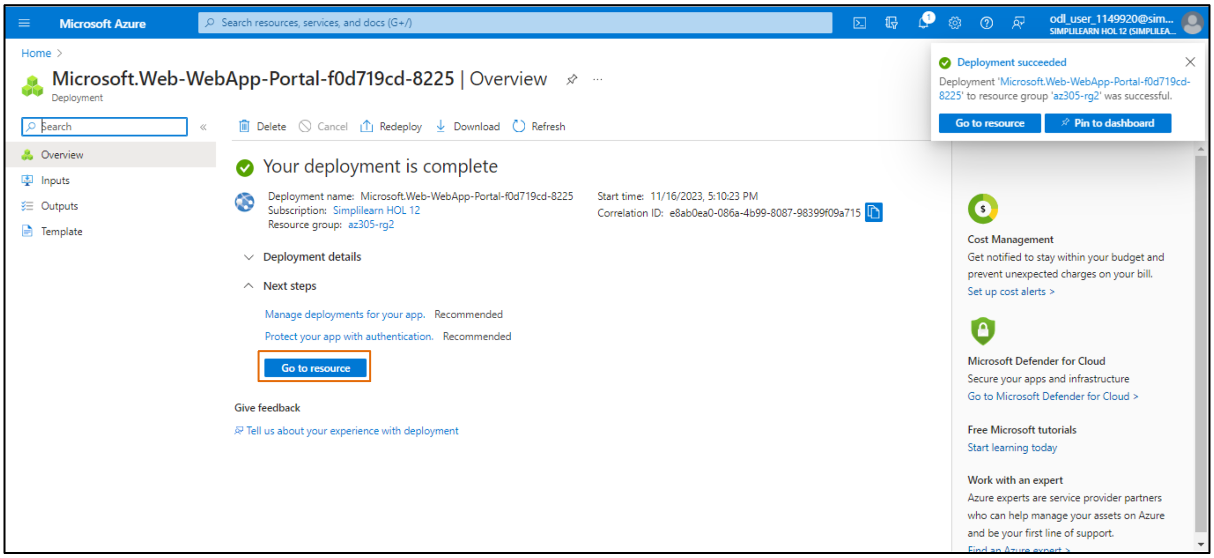
1.4 In the **Review + create** tab, click on **Create**



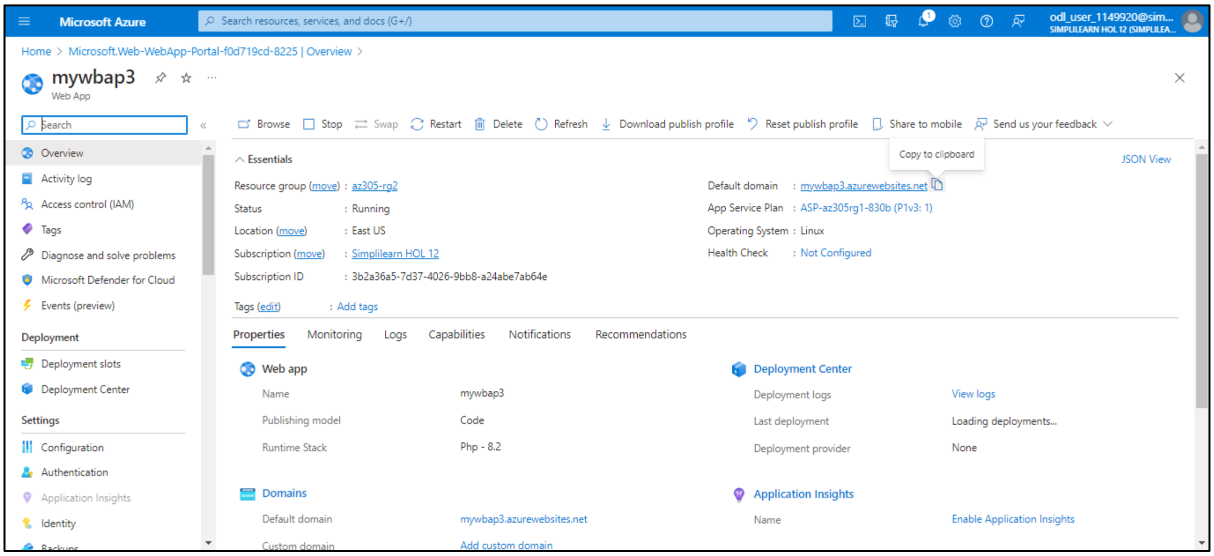
The deployment completion screen appears.



1.5 In the **Overview** page, click on **Go to resource**

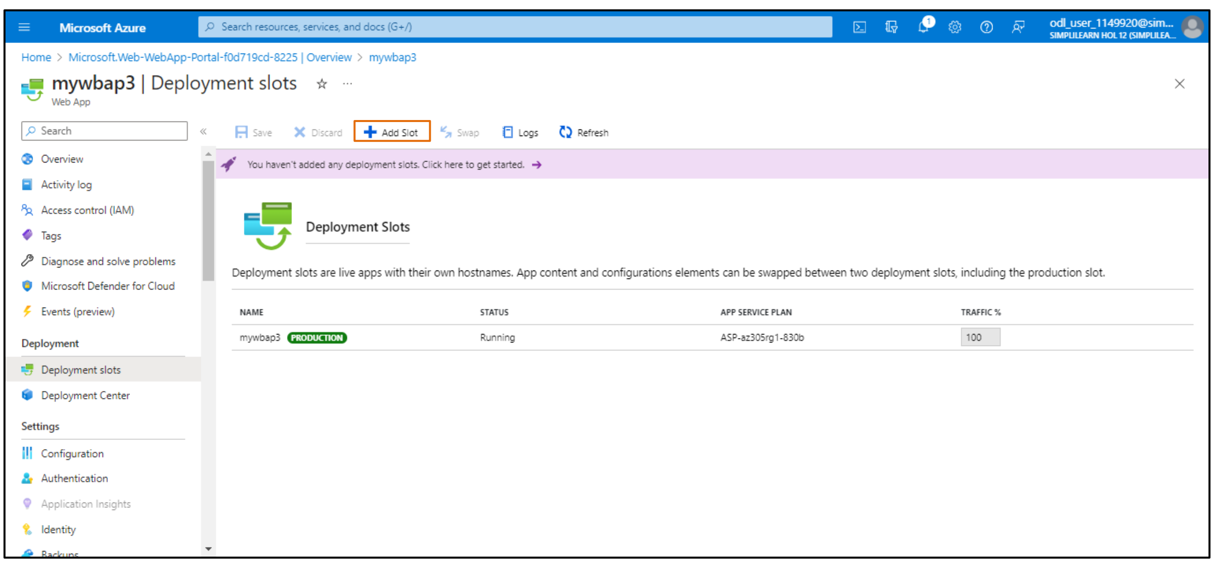


In the newly developed **Web App** page (**mywbap3**), the **Default domain** URL appears:

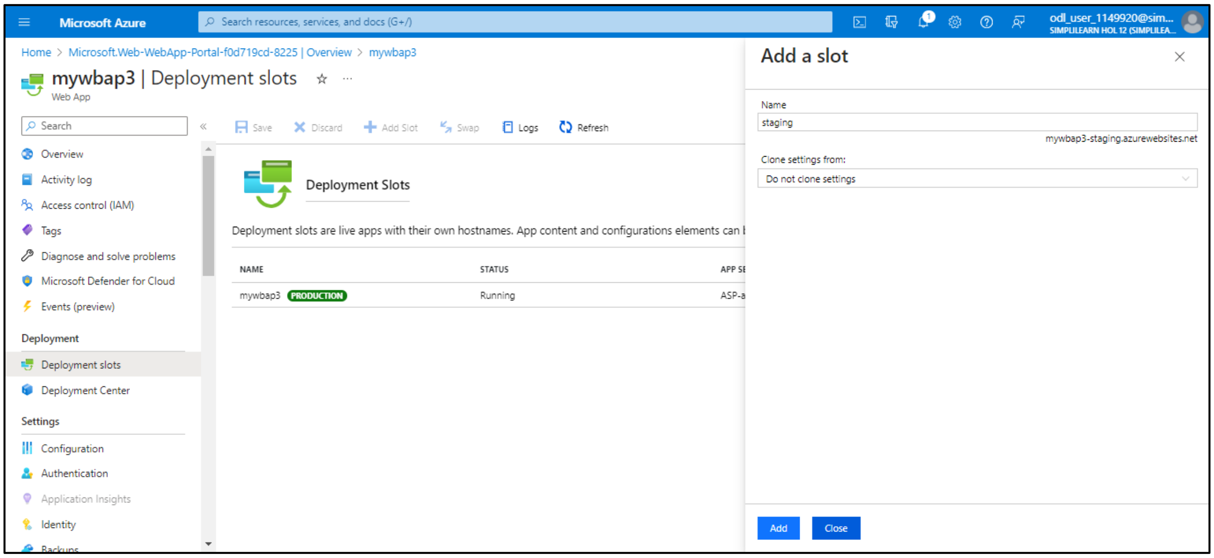


Step 2: Create a staging deployment slot

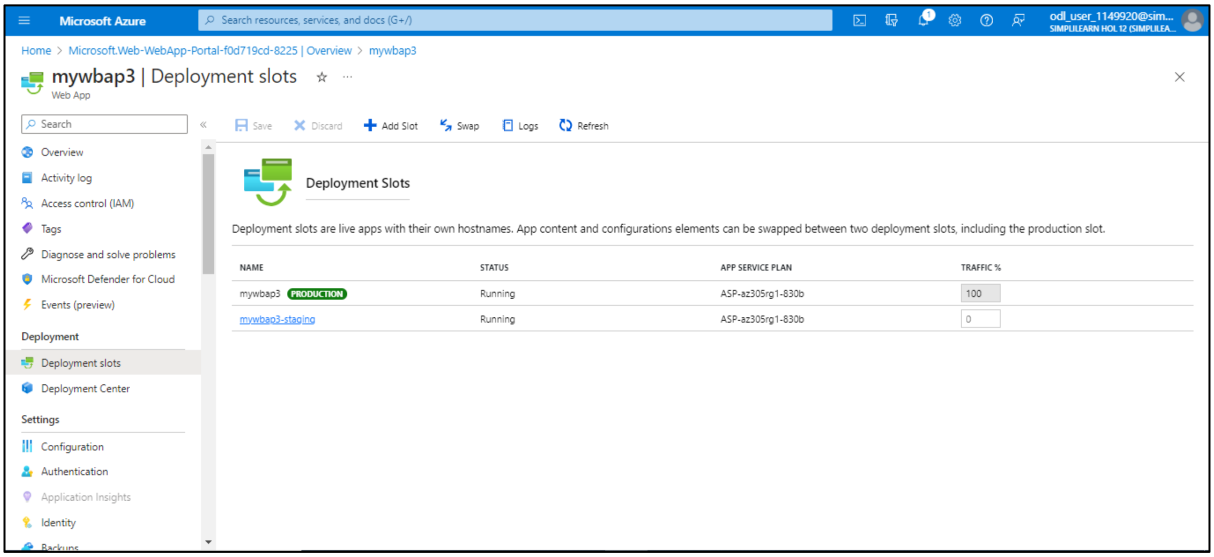
2.1 Under the **Deployment** section of the left side pane, click on **Deployment slots.** Then, select **Add Slot.**



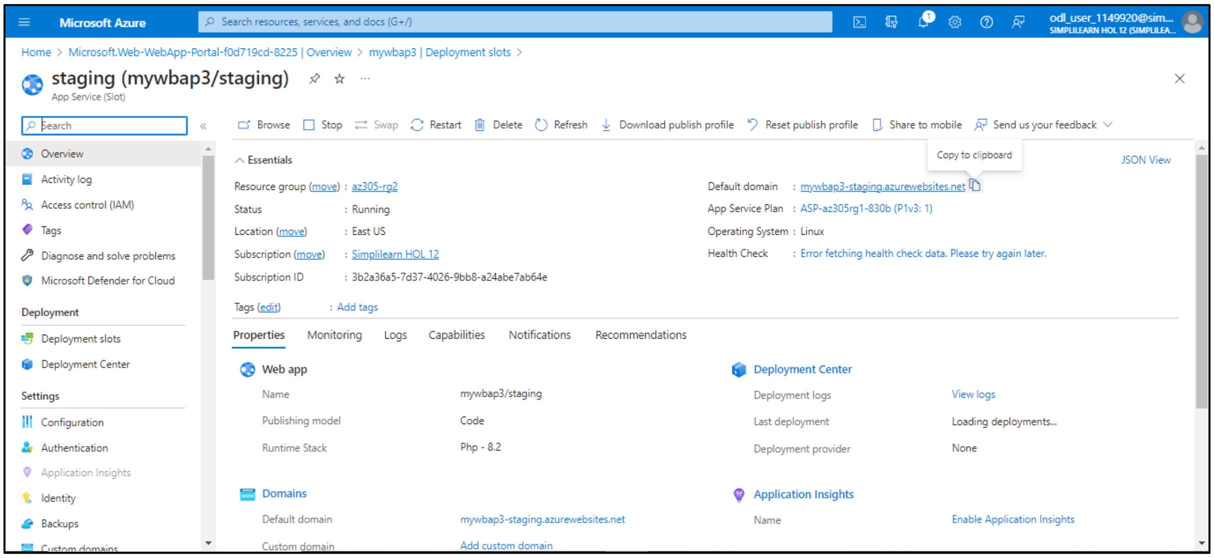
2.2 In the **Add a slot** page, enter the **Name** as **staging** and choose the **Clone settings from** as **Do not clone settings.** Once the details are entered, click **Add** and select **Close**.



2.3 Navigate back to the **Deployment slots** page and click on the entry representing the newly created staging slot (**mywbap3-staging**)

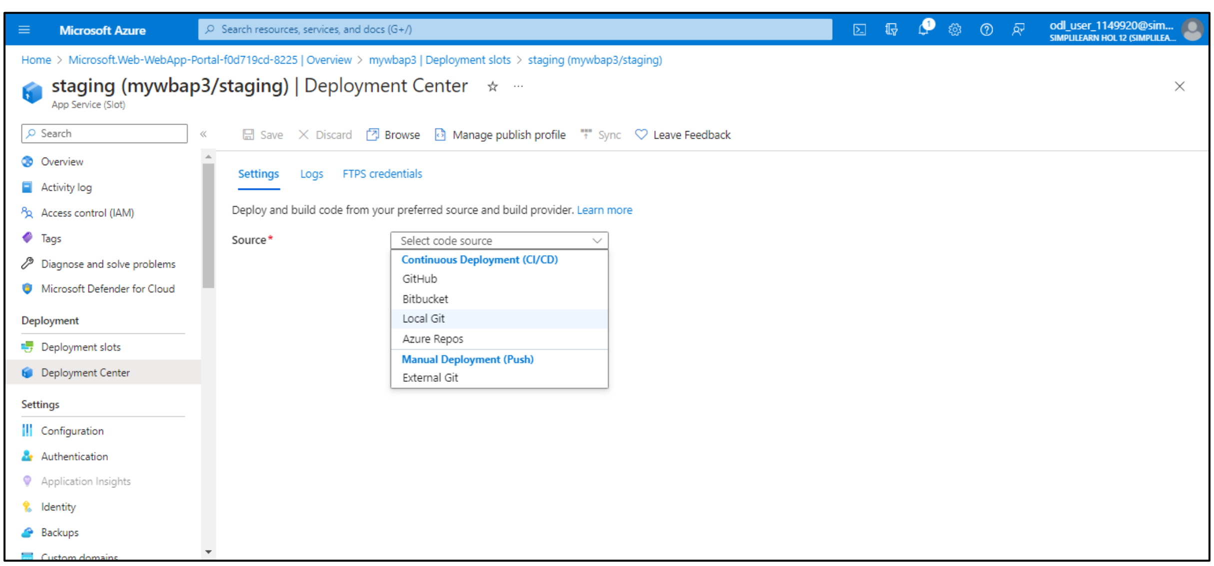


Review the essentials in the **App Service (Slot)** page and note that the **Default domain** URL differs from the one assigned to the production slot.

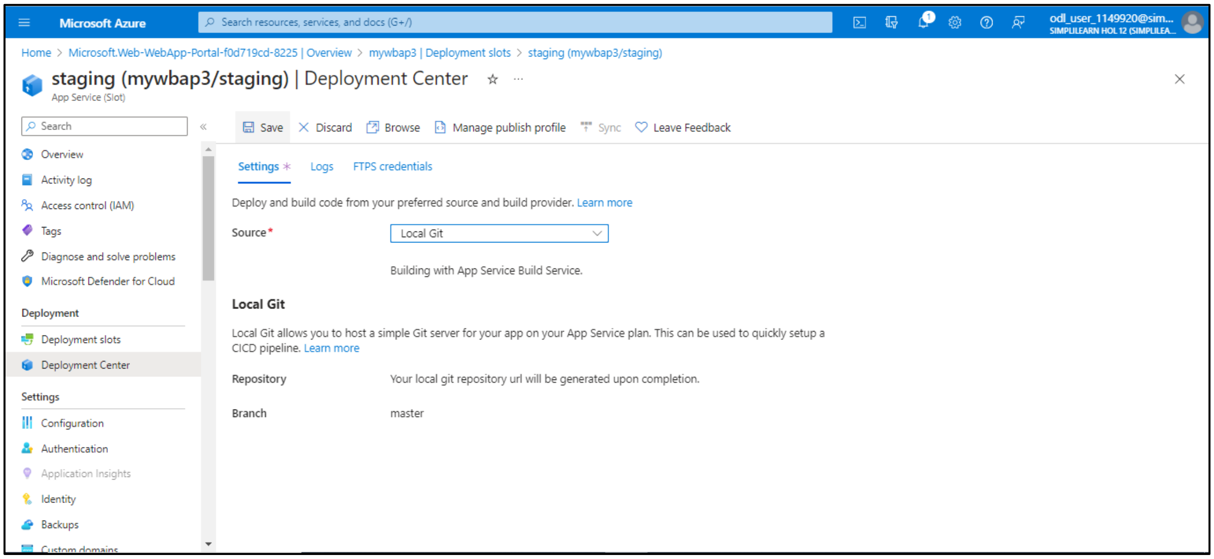


Step 3: Configure web app deployment settings

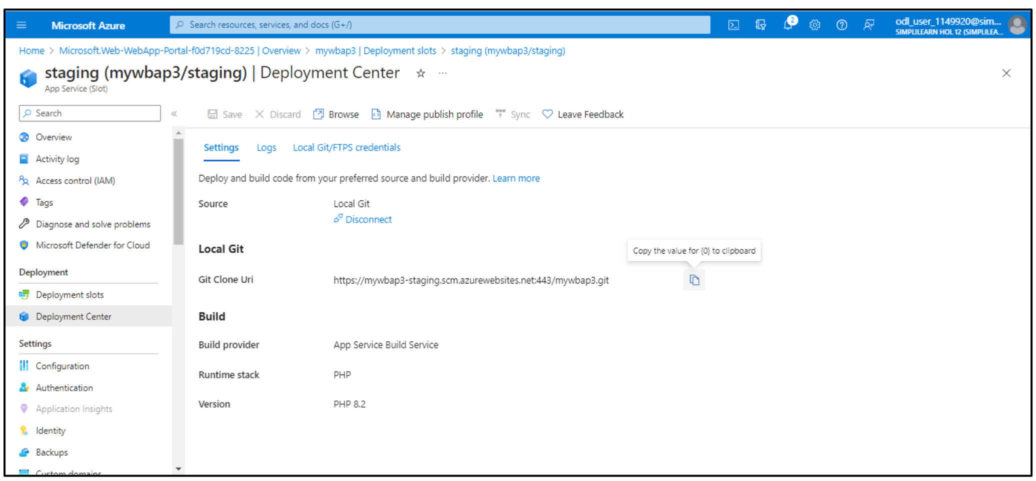
3.1 Under the **Deployment** section, click on **Deployment Center** and then select the **Settings** tab. In the **Source** drop-down list, select **Local Git**.



3.2 Click on **Save**

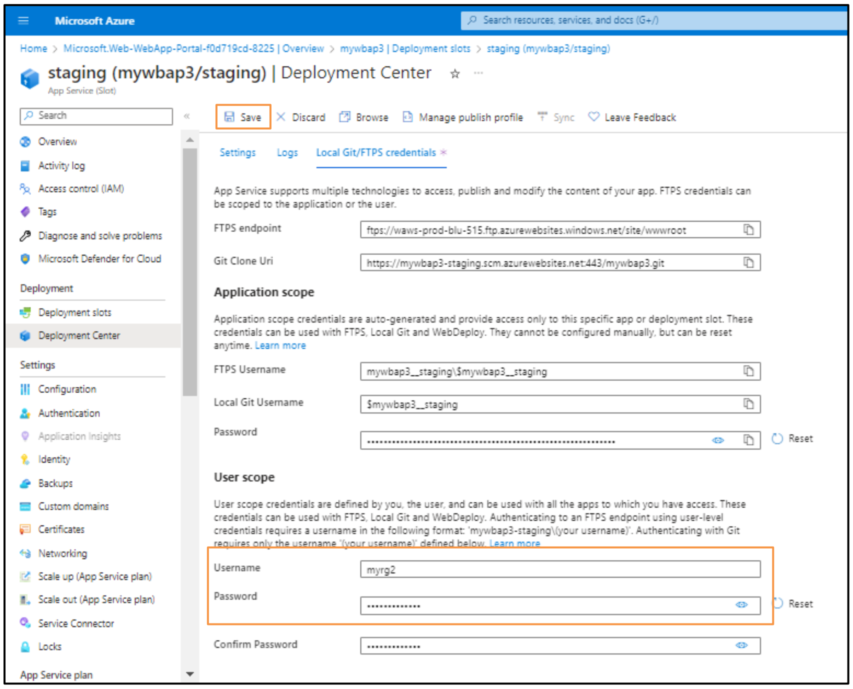


3.3 Under the **Local Git**, copy the **Git Clone Url** to your Notepad



Note: You will need the **Git Clone Url** value in the next task of this lab.

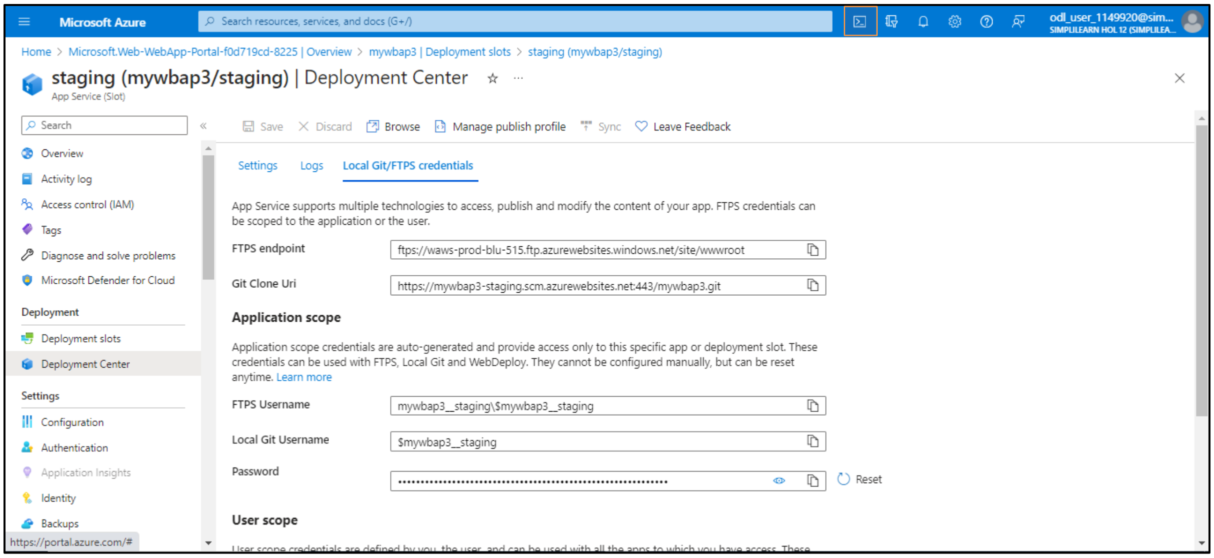
3.4 In the **Deployment Center** page, select the **Local Git/FTPS credentials** tab. Under the **User scope** section, specify the **Username and Password** and click on **Save.**



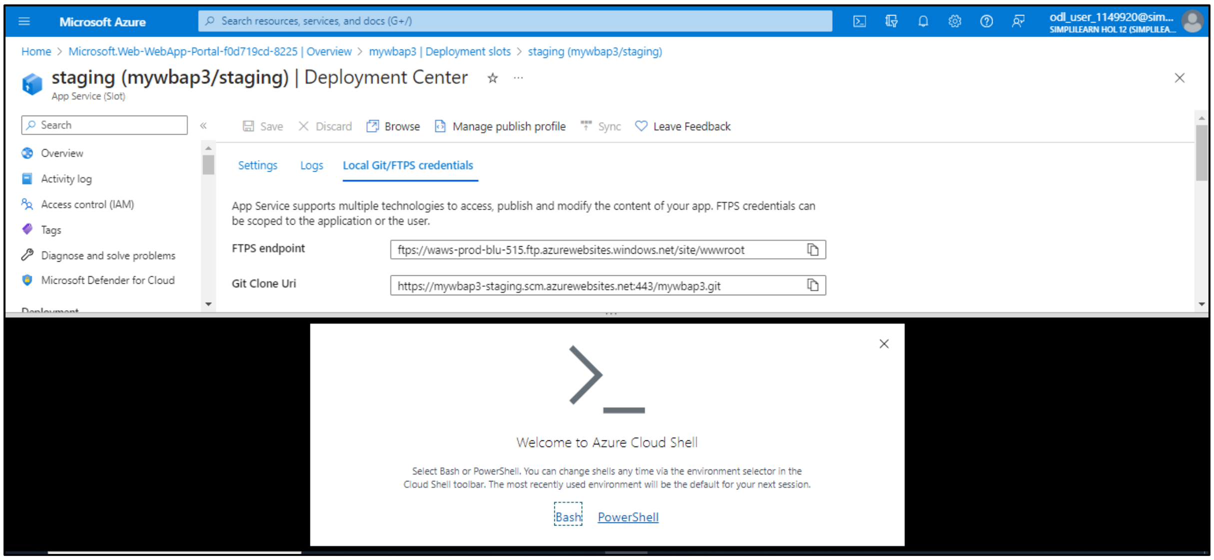
Note: Copy these credentials. Users will need it for the next task in this lab.

Step 4: Deploy code to the staging deployment slot

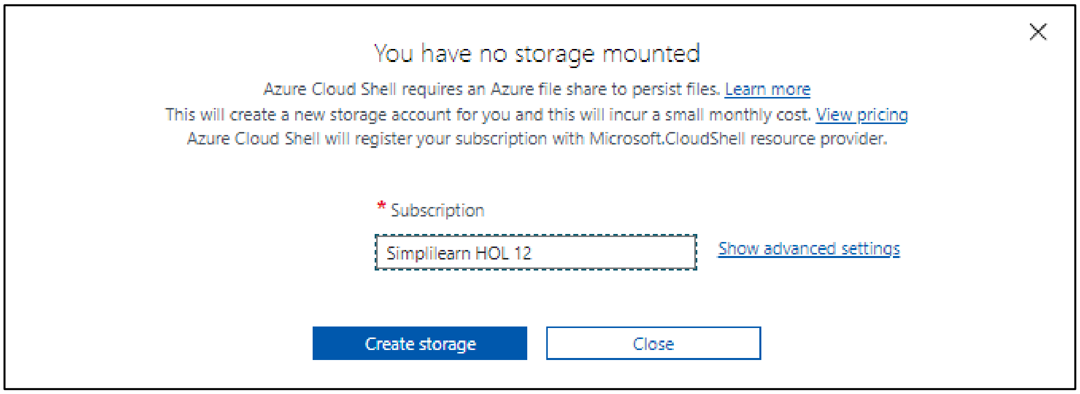
4.1 Click on the **Cloud Shell** icon near the search bar



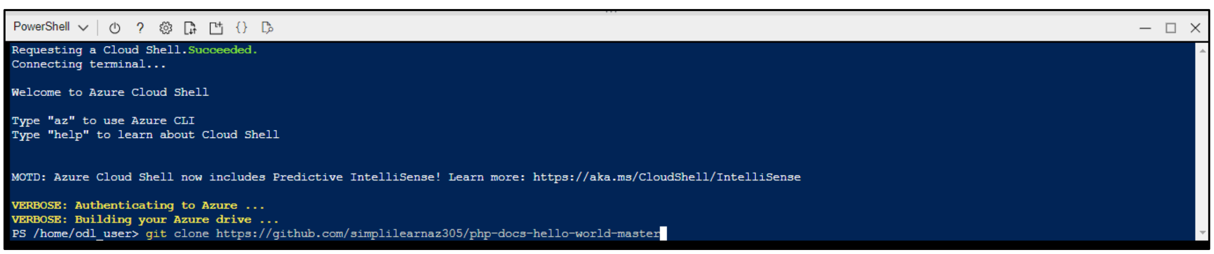
4.2 If prompted to select either **Bash** or **PowerShell**, select **PowerShell**



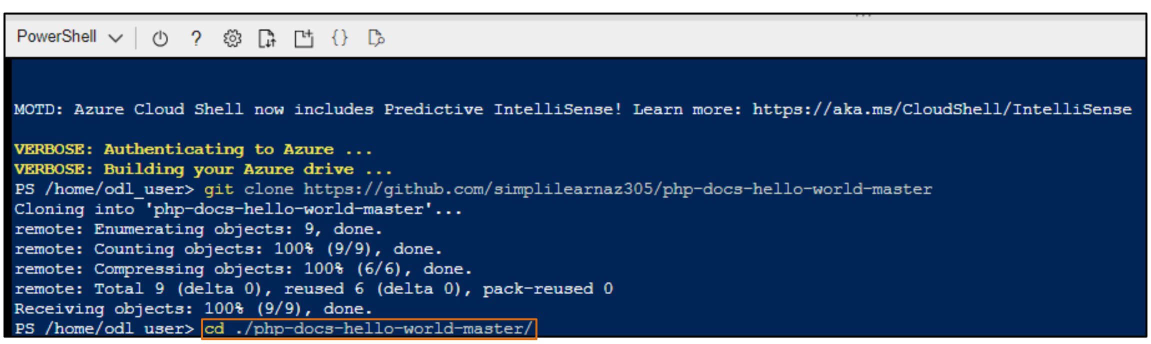
4.3 For first time users, **Cloud Shell** are presented with the **You have no storage mounted** message. Select the subscription you are using in this lab and click on **Create storage**.



4.4 From the **PowerShell** pane, run the following command to clone the remote repository containing the code for the web app:
git clone https://github.com/simplilearnaz305/php-docs-hello-world-master

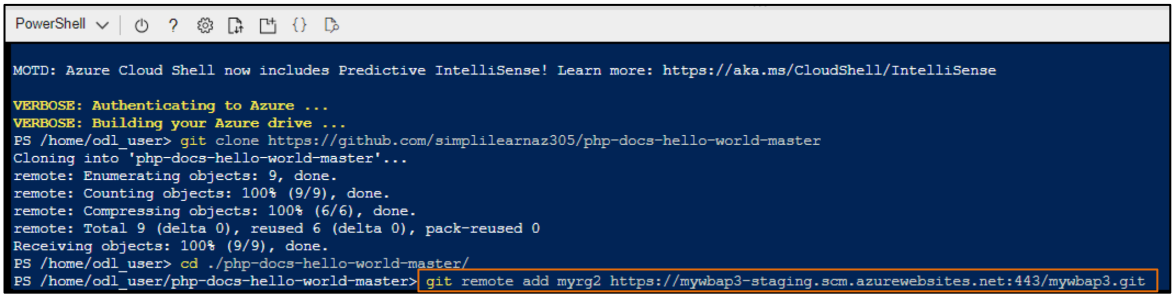


4.5 Run the following command to set the current location to the newly created clone of the local repository:
cd ./php-docs-hello-world-master/



4.6 Run the following command to add the remote git:

git remote add [deployment_user_name] [git_clone_url]



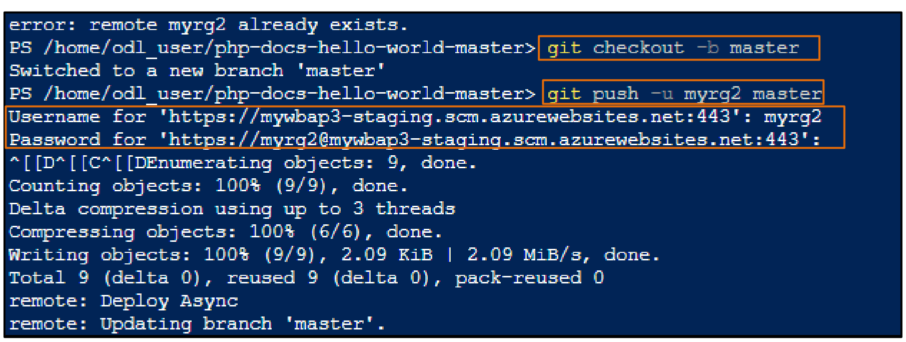
Note: Make sure to replace the [deployment_user_name] and [git_clone_url] placeholders with the value of the **Deployment Credentials** username and **Git Clone Url** respectively, which you identified in the previous task.

4.7 Run the following commands one by one to push the sample web app code from the local repository to the Azure web app staging deployment slot:

git checkout -b master

git push -u [deployment_user_name] master

Type the **[deployment_user_name]** and the corresponding **password** (Set in the previous task)

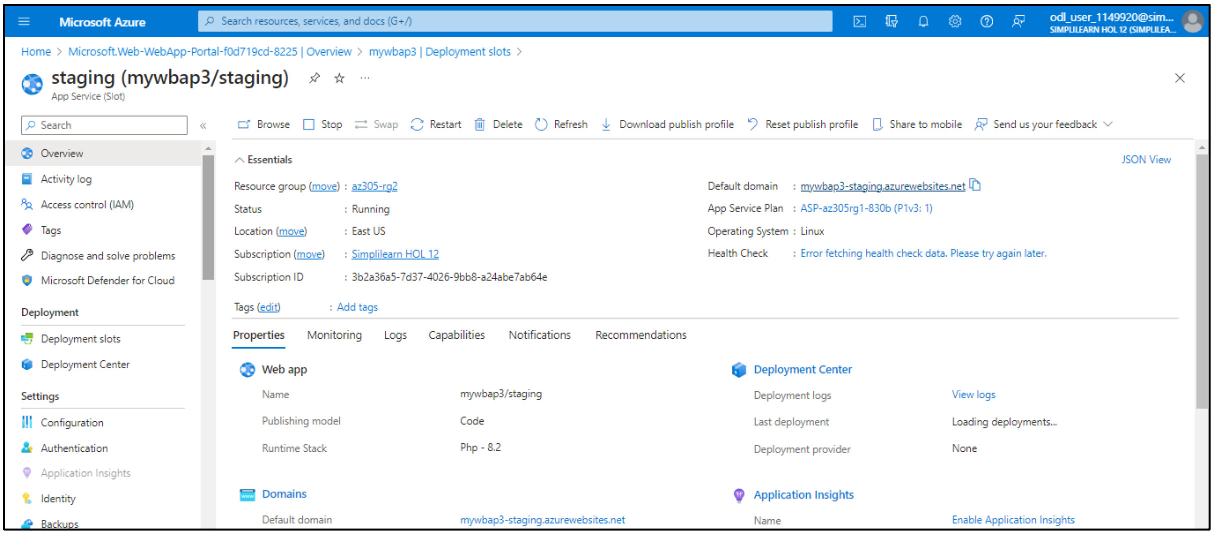


Notes:

1. The value following git remote add does not have to match the **Deployment Credentials** username but must be unique.
2. Make sure to replace the [deployment_user_name] placeholder with the value of the **Deployment Credentials** username, which you identified in the previous task.

The code is deployed as shown in the screenshot below:

4.8 Close the **Cloud Shell** pane. On the **staging** page, select **Overview** and then click on the **Default domain** URL to display the default web page in a new browser tab.



Verify that the browser page displays **Hello World!** message and close the new tab

By following these steps, you have successfully created an Azure web app and a staging deployment slot. You have also configured web app deployment settings and deployed code to the staging deployment slot.